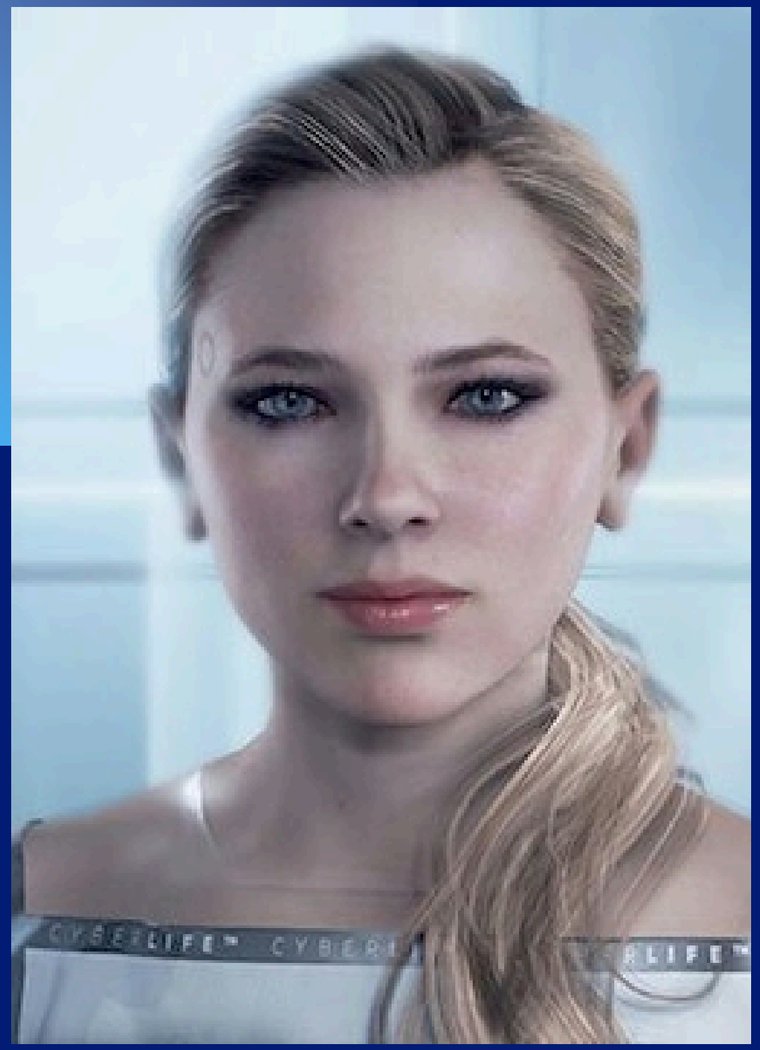
SVCE INNOVATES 2026

SVCE | Sri Venkateswara College of Engineering

Deep Fake Detection SYSTEM

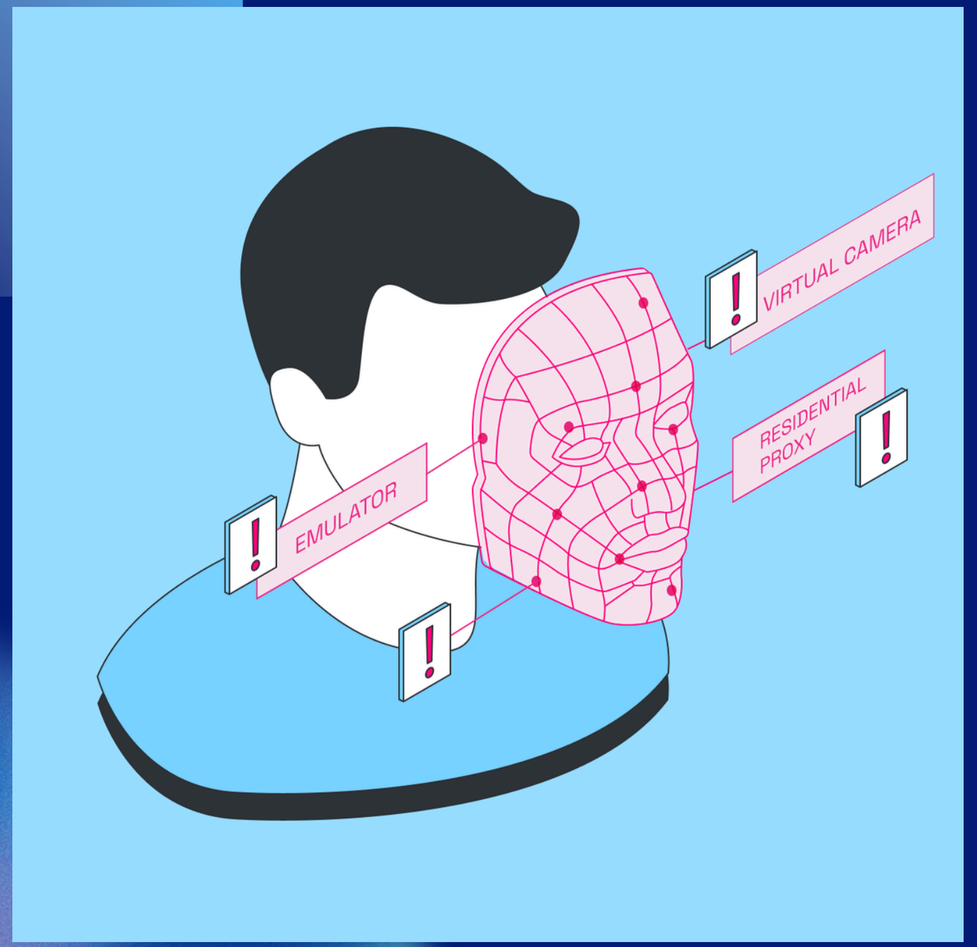


Problem Statement

- Deepfakes are AI-generated fake videos
- Hard to detect with human eyes
- Misused in:
  - Fake news
  - Fraud
  - Political manipulation
- To Design and Develop a Deep Learning algorithm to classify the video as deepfake or pristine

Introduction

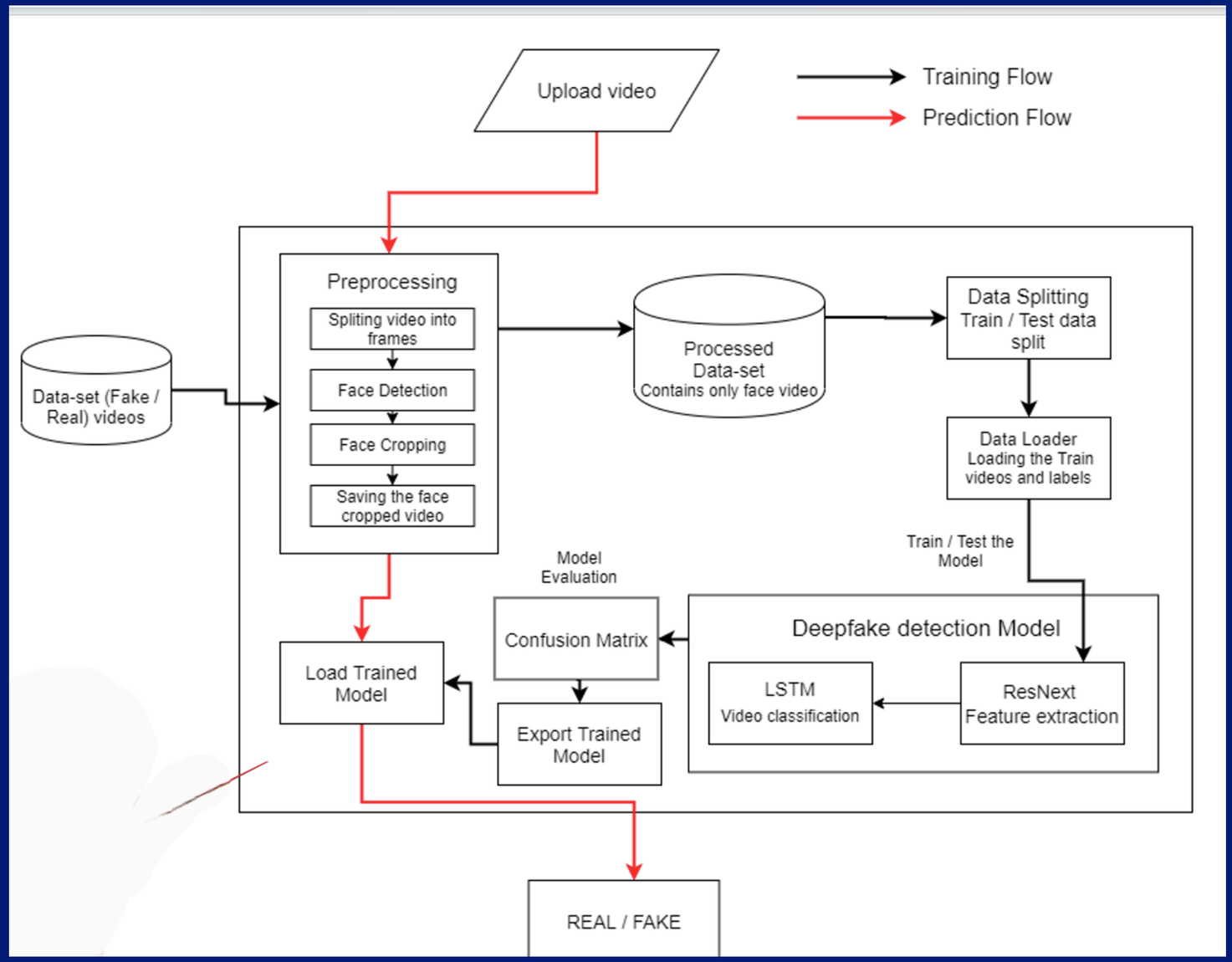
•Deep fake is a technique for human image synthesis based on artificial intelligence.
•Deep fakes are created by combing and superimposing existing images and videos onto source images or videos using a deep learning technique known as generative adversarial network.



Tools for deep fake creation

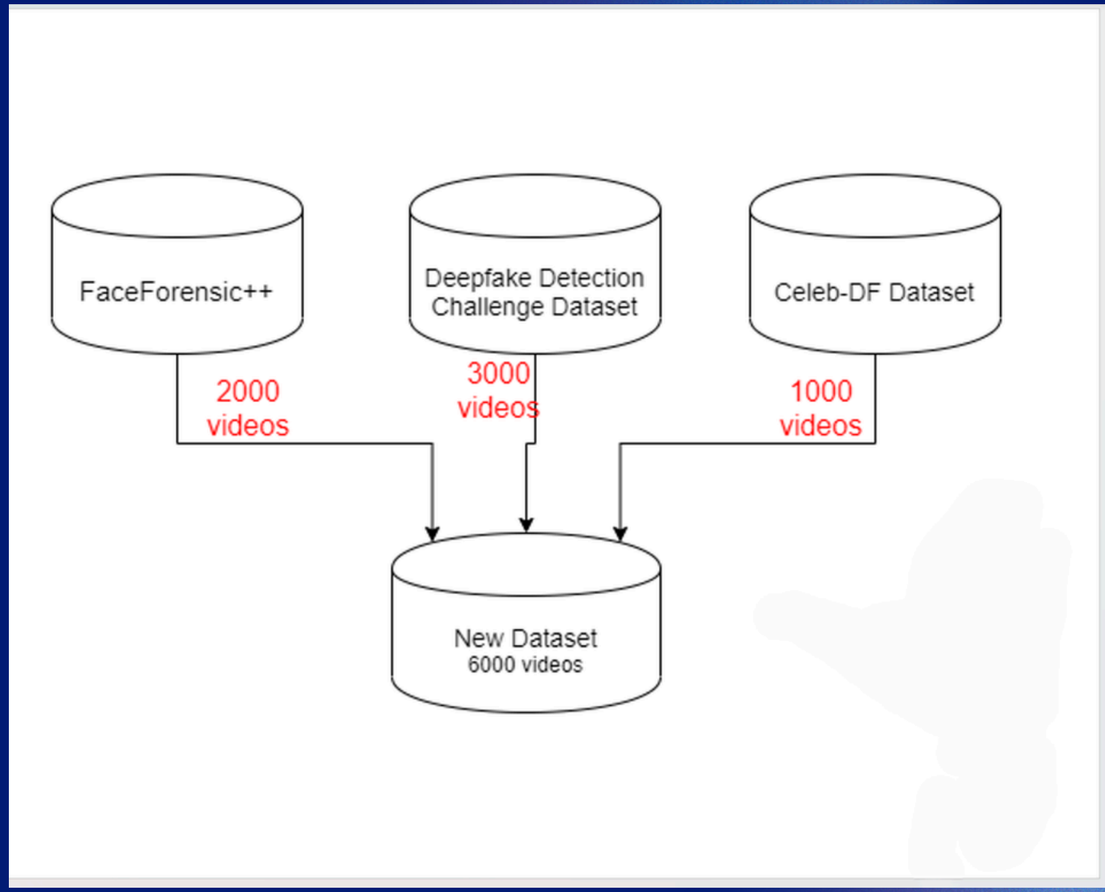
- Faceswap
- Faceit
- DeepFaceLab
- DeepfakeCapsuleGAN
- Large resolution facemasked

System Architecture



Data-set Exploration Architecture

The Dataset Exploration Architecture shows how multiple datasets are collected, analyzed, and combined to form a final training dataset for the deepfake detection model.



Input Datasets

- FaceForensics++~2000 videos
- Real + manipulated videos
- Deepfake Detection Challenge (DFDC)~3000 videos
- Real-world deepfakes
- Celeb-DF~1000 videos
- High-quality celebrity deepfakes

Dataset Integration

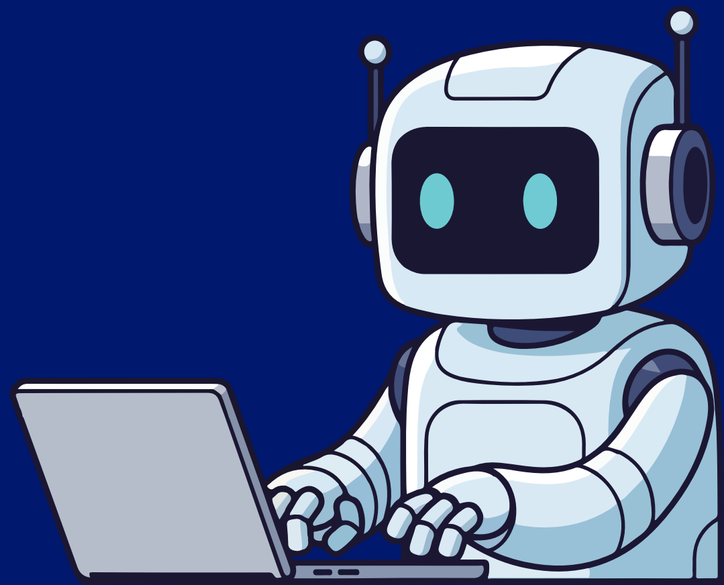
- Merge all datasets
- Total dataset:
  - 6000 videos

Dataset Balancing

- Equal number of:
  - Real videos
  - Fake videos

Pre-processing

1. SPLIT VIDEO INTO FRAMES
2. FACE DETECTION
3. CROPING FACE
4. CREATING NEW FACE CROPPED VIDEO
5. SAVING THE FACE CROPPED VIDEO



Model Architecture

ResNext-50 Encoder

Stage	Output	ResNeXt-50 (32×4d)
conv1	112×112	7×7, 64, stride 2
conv2_x	56×56	3×3 max pool, stride 2 [1×1, 128 3×3, 128, C=32 1×1, 256] ×3
conv3_x	28×28	[1×1, 256 3×3, 256, C=32 1×1, 512] ×4
conv4_x	14×14	[1×1, 512 3×3, 512, C=32 1×1, 1024] ×6
conv5_x	7×7	[1×1, 1024 3×3, 1024, C=32 1×1, 2048] ×3
	1×1	Global Average Pool 1000-d fc, softmax
# params.		25.0×10^6

Sequential Layer

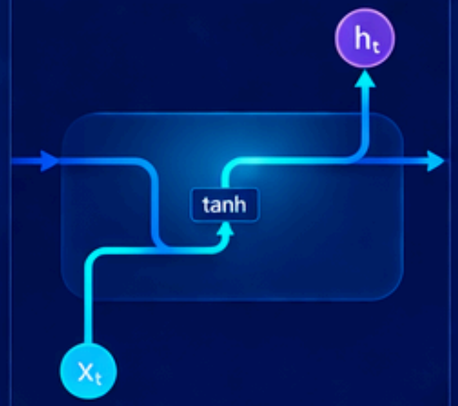
LSTM Classifier 2048

1 LSTM layer processes the **2048-d shape vector**

Captures latent features with sequential modeling

Includes **Dropout (0.4)** and **ReLU activation**

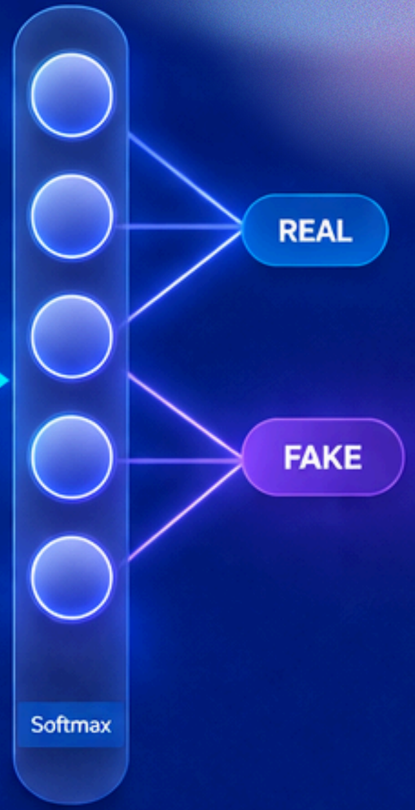
Activation Flow



Hidden Size: 2048 • Dropout: 0.4

Output Layer

Adaptive Average Pooling

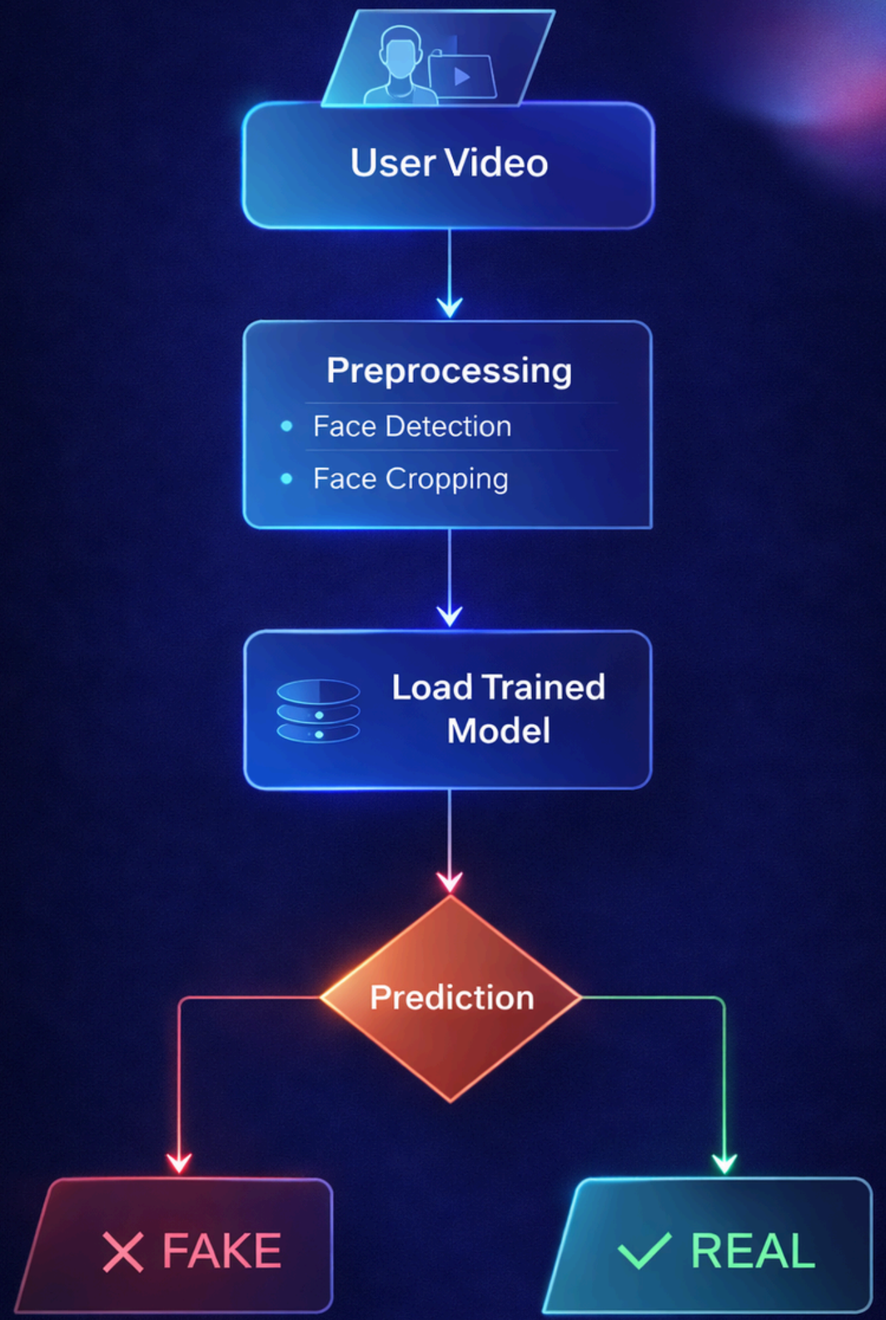


REAL

FAKE

Training Workflow

Prediction Workflow



Programming Languages
- Python3
- JavaScript

JS

Programming Frameworks
- PyTorch
- Django

django

IDE
- Google colab
- Jupyter Notebook
- Visual Studio Code

jupyter

Cloud Services
- Google Cloud Platform

Version Control
- Git

Results

Model Name	Dataset	No of Videos	Sequence Length	Accuracy
model_90_acc_20_frames_FF_data	FaceForensic++	2000	20	90.95477387
model_95_acc_40_frames_FF_data	FaceForensic++	2000	40	95.22613065
model_97_acc_60_frames_FF_data	FaceForensic++	2000	60	97.48743719
model_97_acc_80_frames_FF_data	FaceForensic++	2000	80	97.73366834
model_97_acc_100_frames_FF_data	FaceForensic++	2000	100	97.76180905
model_84_acc_10_frames_final_data	Our Dataset	6000	10	84.662519
model_87_acc_20_frames_final_data	Our Dataset	6000	20	87.79160186
model_89_acc_40_frames_final_data	Our Dataset	6000	40	89.356181195956
model_91_acc_60_frames_final_data	Our Dataset	6000	60	91.590797822706
model_92_acc_80_frames_final_data	Our Dataset	6000	80	92.498185583387
model_93_acc_100_frames_final_data	Our Dataset	6000	80	91.590979782706
model_93_acc_100_frames_final_data	Our Dataset	6000	100	92.108838777

SVCE | Sri Venkateswara College of Engineering

SVCE INNOVATES 2026

References

1.**Rossler et al.,**
“FaceForensics++: Learning to Detect Manipulated Facial Images”, arXiv, 2019.

2.**Deepfake Detection Challenge (DFDC) Dataset,**
Kaggle: https://www.kaggle.com/c/deepfake-detection-challenge/data

3.Li et al.,
“Celeb-DF: A Large-scale Challenging Dataset for DeepFake Forensics”, arXiv, 2019.

4.Güera & Delp,
“Deepfake Video Detection Using Recurrent Neural Networks”, IEEE AVSS, 2018.

5.Li & Lyu,
“Exposing DeepFake Videos by Detecting Face Warping Artifacts”, arXiv, 2018.

6.Li et al.,
“Exposing AI Created Fake Videos by Detecting Eye Blinking”, arXiv, 2018.

7.Nguyen et al.,
“Using Capsule Networks to Detect Forged Images and Videos”, arXiv, 2018

8.**Ciftci et al.,**
“Detection of Synthetic Portrait Videos using Biological Signals”, arXiv, 2019.

9.Kingma & Ba,
“Adam: A Method for Stochastic Optimization”, arXiv, 2014.

10.ResNeXt Model Documentation,
https://pytorch.org/hub/pytorch_vision_resnext/

11.PyTorch Documentation,
https://pytorch.org/

12.Deepfake Video Detection using Neural Networks,
IJSRD Journal Paper

SVCE | Sri Venkateswara College of Engineering

SVCE INNOVATES 2026

THANK YOU

Project by:
BHISHMA KUMAR A
ABIJITH P
BAARATH ARUMUGARAJA

Guided by:
Ajoe Sweetlin Jeena